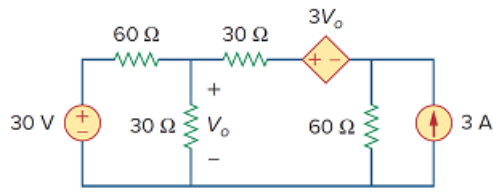


Problem

Use nodal analysis to find V_o in the circuit of Fig. 3.72.

Figure 3.72:

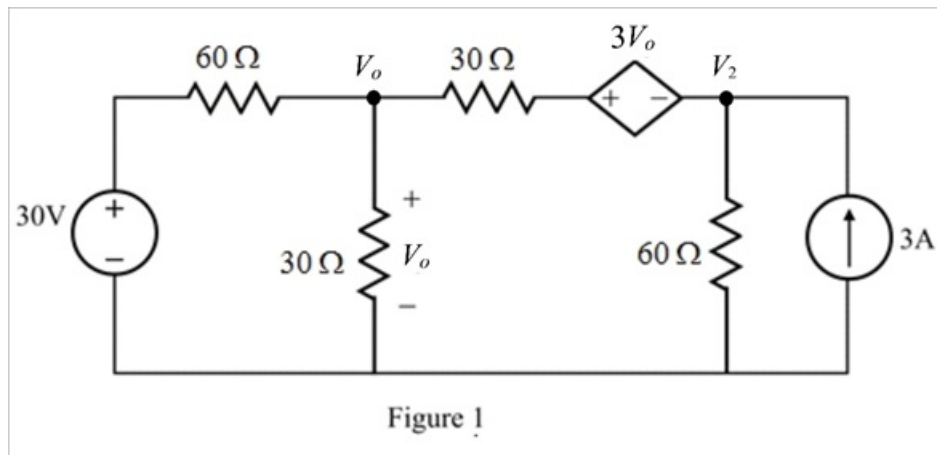


Step-by-step solution

Step 1 of 3

Refer to Figure 3.72 in the textbook.

Redraw the circuit as shown in Figure 1.



Step 2 of 3

Apply Kirchhoff's current law at node V_o .

$$\frac{V_o - 30 \text{ V}}{60 \Omega} + \frac{V_o}{30 \Omega} + \frac{V_o - 3V_o - V_2}{30 \Omega} = 0$$

$$V_o \left[\frac{1}{60} + \frac{1}{30} - \frac{2}{30} \right] - \frac{V_2}{30} = \frac{1}{2}$$

$$V_o \left[\frac{-1}{60} \right] - \frac{V_2}{30} = \frac{1}{2}$$

$$V_o + 2V_2 = -30 \dots\dots (1)$$

Apply Kirchhoff's current law at node V_2 .

$$\frac{V_2 - V_o + 3V_o}{30 \Omega} + \frac{V_2}{60 \Omega} - 3 = 0$$

$$V_2 \left[\frac{1}{30} + \frac{1}{60} \right] + \frac{V_o}{15} = 3$$

$$V_2 \left[\frac{1}{20} \right] + \frac{V_o}{15} = 3$$

$$4V_o + 3V_2 = 180 \dots\dots (2)$$

Write equation (1) and equation (2) in matrix form.

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} V_o \\ V_2 \end{bmatrix} = \begin{bmatrix} -30 \\ 180 \end{bmatrix}$$

Use Cramer's rule to solve equation (1) and equation (2) for voltage V_o .

Determine the value of determinant Δ as follows:

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix} \\ &= (1)(3) - (2)(4) \\ &= 3 - 8 \\ &= -5 \end{aligned}$$

Determine the value of Δ_1 as follows:

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} -30 & 2 \\ 180 & 3 \end{vmatrix} \\ &= (-30)(3) - (2)(180) \\ &= -90 - 360 \\ &= -450 \end{aligned}$$

Determine the voltage V_o .

$$V_o = \frac{\Delta_1}{\Delta}$$

Substitute -5 for Δ and -450 for Δ_1 .

$$\begin{aligned} V_o &= \frac{-450}{-5} \\ &= 90 \text{ V} \end{aligned}$$

Hence, the voltage V_o is 90 V.